

Global Power Plant Analysis
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The Global Power Plant Database is an open-source open-access dataset of grid-scale (1 MW and greater) electricity generating facilities operating across the world. The goal of this analysis is to give an overview of where the power plants are located, how the electricity generation has changed over time and by primary fuel type. In addition, comparison of utilization by primary fuel is done.

This document gives an overview of the analysis process. The analysis itself is visible in the file "Global Electricity Generation Report.pbix".

Questions for the customer

In the first meeting, I would ask what decisions the customer wants to make based on this data, e.g. are they looking to build a new power plant and want to know which type of fuel might be the best. I would additionally ask which measures and dimensions would they like to see in the report.

Summary of the business requirements for starting the report development

I took the questions specified in the task as for what should be shown in the report:

- What types of primary fuel are used to generate energy?
- How generation of electricity is changed year-over-year?
- What is the powerplant Utilisation coefficient for different fuel types?
- Where are powerplants located on the map?
- What insights can be gained from the analysis of temporal trends in the data? (e. g. global trend of fuel type market share over time)
- What is the average lifetime of powerplant?

Based on the questions, I decided to create two report pages: power plants overview and electricity generation and utilization overview.

The first page gives a view of where the power plants are located and details about each plant. It is possible to filter out plants based on if they had electricity generation data on a specific year and which type of primary fuel the plant uses.

The second page gives an overview of electricity generation by year, electricity generation and average utilization % by primary fuel and how electricity generation has changed over time by primary fuel type.

The question of average lifetime of a powerplant would need clarifying on how it should be calculated so this is not presented in the report.

The report is meant for general public who wants to have an overview of how the power plant market looks like and how has it changed over the years.

Data quality controls

I used Power Query to import the data and then Column Profiling tools to check for data quality issues. Each column was checked separately and the main data quality questions that arised:

- 1) Can the energy generation be a negative number?
- 2) Can the energy generation be missing for many plants?

As the energy generation can be negative in cases the maintenance of plant takes more energy than the plant produces then the negative numbers were not removed.

Missing energy generation data can come from two reasons:

- 1) The plant has closed down.
- 2) The data for that plant was not entered into the database for that specific year.

As there is no further info available on this, no changes were made to the data. It would be possible to take the estimate but it would require a definite knowledge that the specific plant was operating that year.

Usage of the report

The Power BI report can be opened with Microsoft's free software Power BI Desktop (downloadable at: <https://www.microsoft.com/en-us/download/details.aspx?id=58494>)

The report includes four visible pages:

- 1) Introduction
 - Gives an overview of the report and dataset.
- 2) Power Plants Overview
 - Gives an overview of the power plants and details of each power plant.
- 3) Electricity Generation
 - Gives an overview of electricity generation by year, electricity generation and average utilization % by primary fuel and how electricity generation has changed over time by primary fuel type.
 - By clicking on a specific primary fuel type in the bar charts, the bottom chart will be filtered to show only the selected primary fuel changes. This enables us to see changes of all fuel types over time (even the smaller amounts). For example, we could see that Solar and Wind has increased over time.
- 4) Conclusion
 - Gives summary of main findings and further research topics.